

the first negative half-cycle, C1 is charged to $V_{RF} - V_T$.

When the positive half-cycle starts, D1 is reverse-biased and the right-end plate of C1 is pushed to $2^*(V_{RF} - V_T)$. D2 is turned on, and charges are transferred to C2. At the end of the positive half-cycle, C2 is charged to $2^*(V_{RF} - V_T)$, for a rectifier with more stage doublers.

Differential-drive bridge rectifier. A full-wave bridge rectifier, as shown in Fig. 4, has been commonly used in AC-DC voltage conversion. The rectifier has a differential input RF signal.

In the positive half-cycle of an RF signal, diodes D2 and D3 conduct, while D1 and D4 are reverse-biased.

In the negative half-cycle of the RF signal, diodes D1 and D4 conduct, while D2 and D3 are reverse-biased.

In the whole cycle, load capacitor C1 is charged in a single direction. DC voltage on C1 can achieve $V_{RF} - (2^*V_{th})$, when the reverse leakage current and other resistive load are considered, where V_{th} is the threshold voltage of the diodes.

The circuit starts rectifying once amplitude of the input AC signal becomes larger than twice the threshold voltage of the diode, which reduces the voltage sensitivity of the rectifier. To improve voltage sensitivity, a gate-drain-connected n-type metal oxide semiconductor field effect transistor (MOSFET) or gate-source connected p-type MOSFET with low-threshold voltage can be used instead of the diode, as shown in Fig. 6(b). To increase DC voltage level, a stack architecture is applicable by cascading the single-cell bridge rectifier in stages by employing coupling capacitors for DC blocking.

Gate cross-connected differential-drive bridge rectifier. Although by using the diode-connected MOSFET with low-threshold voltage, voltage sensitivity of the bridge rectifier can be effectively

TABLE II
Pros and Cons of MOSFET and Schottky Diode

Device	Pros	Cons
MOSFETs	Compatible with standard EEPROM processes Low threshold voltage <0.1V	Higher substrate loss
Schottky diodes	Not compatible with standard EEPROM processes	Low substrate loss

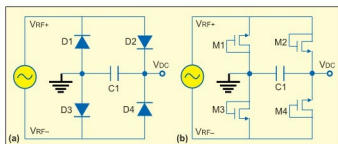


Fig. 6: Full-wave bridge rectifier

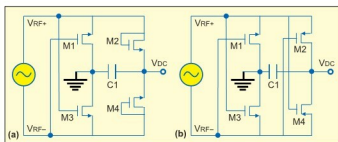


Fig. 7: Differential drive bridge

improved, reverse leakage power consumption induced by the MOSFET is not negligible.

An adapted architecture of the conventional bridge rectifier is to bias the gate terminal with the differential input RF signal. By biasing the gate terminal, turn-on voltage of the MOSFET is reduced, with which voltage sensitivity can be effectively improved.

Consider the architecture in Fig. 7(a) as an example. In the positive half-cycle of the RF signal, as in a normal bridge rectifier, M2 and M3 conduct, while M1 and M4 are reverse-biased. The change happens in the bias voltage in the gate terminal of M1 and M3.

In the positive half-cycle, M3 conducts with a bias signal larger than zero (ground potential) as in a conventional bridge rectifier, which reduces threshold voltage of M3 and hence improves voltage sensitivity.

In the positive half-cycle, drain

and source terminals of M1 are exchanged. M1 is reverse-biased at the gate terminal with the negative RF signal, which is lower than the ground potential of the source terminal. With such a bias, leakage current induced by M1 is largely reduced.

The same analysis can be done for the negative half-cycle. In an NMOS gate cross-connected bridge rectifier, only two MOSFETs are biased differentially.

To further improve the performance of the rectifier, two PMOS devices are used instead, as shown in Fig. 7(b). In the positive half-cycle, PM2 and NM3 are conducted, while NM1 and PM4 are reverse-biased. Bias signal at the gate terminal of PM2 is negative, which results in a smaller turn-on voltage than the diode-connected mode. Source and drain terminals of PM4 exchange. Gate terminal of PM4 is biased with the positive RF signal, which is larger than the potential at its source terminal and hence reduces the reverse leakage current. The same analysis can be done for the negative half-cycle.

Commonly-used devices for rectifier design are MOSFETs and Schottky diodes. The difference in physical construction results in the disadvantages and advantages in the application of an RF-DC rectifier design. Table II summarizes the pros and cons of both devices. ●